

CITYCARE DIAGNOSTIC LABORATORY

Clinical Pathology | Hematology | Biochemistry

LABORATORY REPORT

SAMPLE / SYNTHETIC REPORT - FOR COLLEGE HACKATHON AND SOFTWARE TESTING

Report No.	DEMO-2026-1003	Patient ID	DEMO-P001
Patient Name	Aarav Shah	Age / Sex	24 Years / Male
Collected	03-Oct-2026 08:15 AM	Sample	Blood / Serum
Referring Doctor	Dr. Demo Physician	Report Status	Final - Sample

CBC / HEMATOLOGY

Test	Result	Reference Range	Flag
Hemoglobin	15.1 g/dL	13.0 - 17.0	NORMAL
RBC	5.10 million/uL	4.5 - 5.9	NORMAL
WBC	6,800 /uL	4,000 - 11,000	NORMAL
Platelets	265,000 /uL	150,000 - 450,000	NORMAL
Hematocrit	44.2 %	40 - 52	NORMAL
MCV	87 fL	80 - 100	NORMAL
MCH	29.6 pg	27 - 33	NORMAL
MCHC	34.2 g/dL	32 - 36	NORMAL
RDW	13.1 %	11.5 - 14.5	NORMAL

DIABETES / GLUCOSE

Test	Result	Reference Range	Flag
Fasting Glucose	89 mg/dL	70 - 99	NORMAL
HbA1c	5.2 %	Below 5.7	NORMAL
Estimated Average Glucose	103 mg/dL	Derived value	NORMAL

RENAL / KIDNEY

Test	Result	Reference Range	Flag
BUN	14 mg/dL	6 - 20	NORMAL
Creatinine	0.9 mg/dL	0.6 - 1.3	NORMAL
Uric Acid	5.4 mg/dL	3.4 - 7.0	NORMAL
eGFR	Above 90 mL/min/1.73m2	90 or above	NORMAL

LIVER

Test	Result	Reference Range	Flag
ALT	22 U/L	4 - 36	NORMAL
AST	20 U/L	8 - 33	NORMAL
ALP	78 U/L	20 - 130	NORMAL
Total Bilirubin	0.7 mg/dL	0.1 - 1.2	NORMAL

Test	Result	Reference Range	Flag
Direct Bilirubin	0.2 mg/dL	0.0 - 0.3	NORMAL
Total Protein	7.2 g/dL	6.0 - 8.3	NORMAL
Albumin	4.5 g/dL	3.4 - 5.4	NORMAL

LIPID PROFILE

Test	Result	Reference Range	Flag
Total Cholesterol	168 mg/dL	Below 200	NORMAL
LDL	94 mg/dL	Below 100	NORMAL
HDL	52 mg/dL	40 or above (male)	NORMAL
Triglycerides	108 mg/dL	Below 150	NORMAL
VLDL	22 mg/dL	5 - 40	NORMAL

ELECTROLYTES

Test	Result	Reference Range	Flag
Sodium	140 mmol/L	135 - 145	NORMAL
Potassium	4.2 mmol/L	3.7 - 5.2	NORMAL
Calcium	9.4 mg/dL	8.5 - 10.2	NORMAL

Laboratory Remarks: All values are illustrative synthetic values for software testing. Reference intervals may vary by laboratory and method.

DISEASE INTERPRETATION AND ABNORMAL-VALUE GUIDE

HIGH = commonly increased. LOW = commonly decreased. This is a testing reference, not a diagnosis.

1. DIABETES PATTERN

Parameter	Reference	Typical finding	Why it changes
Fasting glucose	70 - 99 mg/dL	HIGH: 126 mg/dL or more*	Reduced insulin action can leave more glucose in the blood.
HbA1c	Below 5.7 %	HIGH: 6.5 % or more*	Shows average blood glucose over roughly the previous 2 - 3 months.
Estimated average glucose	Based on HbA1c	HIGH when HbA1c is high	Calculated estimate of average glucose.

2. ANEMIA PATTERN

Parameter	Reference	Typical finding	Why it changes
Hemoglobin	Male about 13 - 17 g/dL	LOW	Reduced oxygen-carrying capacity; causes include iron, B12 or folate deficiency and blood loss.
Hematocrit	Male about 40 - 52 %	LOW	Lower proportion of blood volume is made up of red blood cells.
MCV	80 - 100 fL	LOW or HIGH	LOW often gives a microcytic pattern; HIGH gives a macrocytic pattern.

3. KIDNEY / RENAL PATTERN

Parameter	Reference	Typical finding	Why it changes
Creatinine	About 0.6 - 1.3 mg/dL	HIGH	May rise when kidney filtration is reduced; interpretation also depends on muscle mass.
BUN	About 6 - 20 mg/dL	HIGH	Can rise with reduced kidney clearance, dehydration and other causes.
eGFR	90 or above	LOW, especially persistent below 60	Estimates kidney filtration; persistent reduction may indicate chronic kidney disease.

4. LIVER INJURY PATTERN

Parameter	Reference	Typical finding	Why it changes
ALT	About 4 - 36 U/L	HIGH	Can be released when liver cells are injured.
AST	About 8 - 33 U/L	HIGH	Can rise with liver injury and also with muscle or other tissue injury.
Bilirubin	About 0.1 - 1.2 mg/dL	HIGH	May rise when bilirubin production, processing or excretion is disturbed.
Albumin	About 3.4 - 5.4 g/dL	LOW in some chronic liver disease	Can fall with reduced production in significant chronic liver dysfunction, but has other causes too.

5. DYSLIPIDEMIA PATTERN

Parameter	Desirable	Typical finding	Why it changes
Total cholesterol	Below 200 mg/dL	HIGH: 200 or more	Higher levels can be associated with increased long-term cardiovascular risk.
LDL	Below 100 mg/dL	HIGH: 100 or more	Elevated LDL is associated with atherosclerotic cardiovascular risk.
HDL	40 mg/dL or above (male)	LOW: below 40	Low HDL is associated with cardiovascular risk.
Triglycerides	Below 150 mg/dL	HIGH: 150 or more	Can rise due to metabolic, dietary and genetic factors.

6. ELECTROLYTE ABNORMALITIES

Parameter	Reference	Typical finding	Why it changes
Sodium	135 - 145 mmol/L	HIGH or LOW	Often related to water balance, kidney function, hormones or fluid losses.
Potassium	3.7 - 5.2 mmol/L	HIGH or LOW	Affected by kidney function, medicines, hormones and movement between cells and blood.
Calcium	8.5 - 10.2 mg/dL	HIGH or LOW	Regulated by parathyroid hormone, vitamin D, kidneys and bone metabolism.

AI PROJECT EXAMPLE

If a synthetic report has fasting glucose 154 mg/dL and HbA1c 7.2 %, the system can flag a diabetes-pattern result. It should not claim a definitive diagnosis from a single report without clinical confirmation.

Important: These are educational synthetic patterns. Real diagnosis depends on symptoms, history, repeat or confirmatory testing and professional clinical interpretation. Laboratory reference ranges may differ.

SAMPLE / SYNTHETIC - NOT A REAL MEDICAL DOCUMENT